# Linkit Kim/SMD

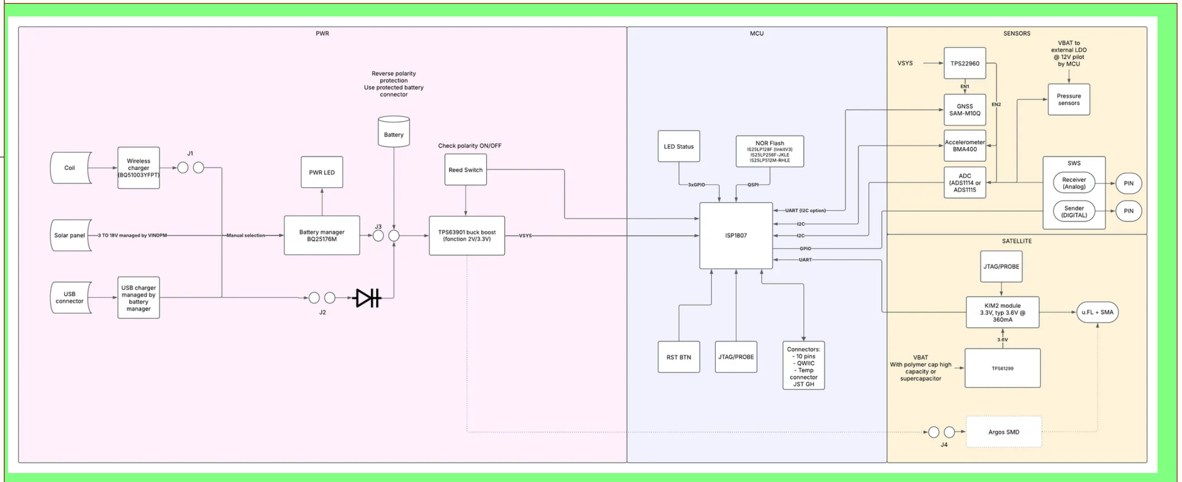
## linkit-v4

Low-power wildlife telemetry platform designed for long-term field deployment.
This variant integrates the SMD Argos module, optimized power management (wireless charging, solar input, and USB/Li-ion handling), and a compact architecture prioritizing energy efficiency, cost, and reliability. All interfaces and components are arranged to ensure seamless compatibility with the Linkit firmware ecosystem and Arribada's open-source tooling.

## Table of contents



U_linkit-v4
linkit-v4.SchDoc




## Legend

General comment such as function title, configuration, ...

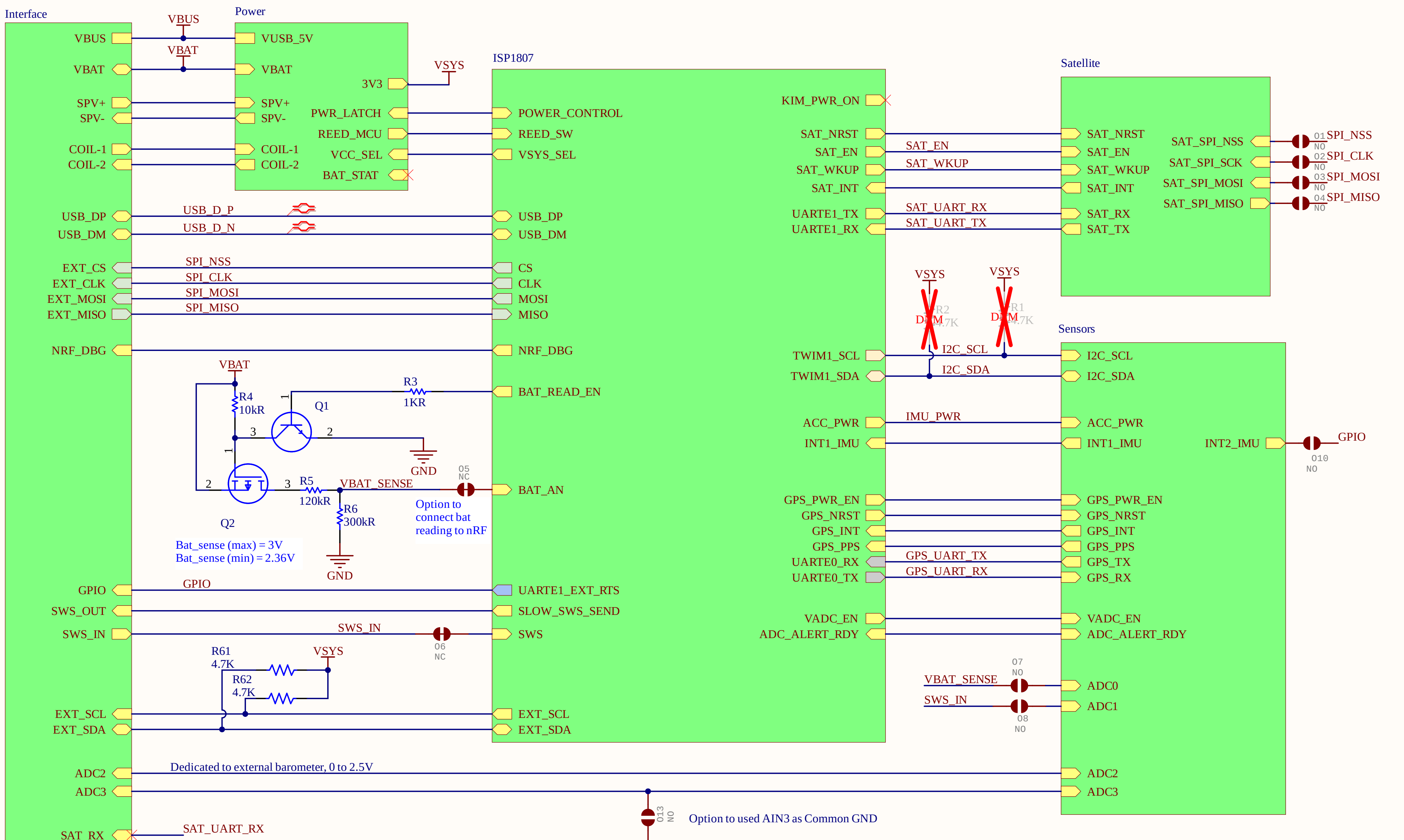
Text to be added to silkscreen.

Warning text.

Notes to generate the board layout.

## License

| | | | |
|---|---|---|---|
| Title **Linkit Kim/SMD** | | | *Arribada Initiative*<br>*https://github.com/arribada*<br>*contact@arribada.org*<br>*arribada.org*<br>*UK - Portsmouth* |
| Size: A4 | Reference: linkit-v4 | Revision: 1 0 | |
| Date: 29/04/2026 | Time: 15:26:31 | Sheet 1 of 12 | |
| File: C:\Users\fourn\Altium\AD25\Projects\Arribada\linkit-v4-hw\Project_Overview.SchDoc | | | |

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| Title | *Linkit Kim/SMD* | | |
| Size: A4 | Reference: linkit-v4 | Revision: 1 0 | |
| Date: 29/04/2026 | Time: 15:26:31 | Sheet 2 of 12 | |
| File: C:\Users\fourn\Altium\AD25\Projects\Arribada\linkit-v4-hw\linkit-v4.SchDoc | | | |


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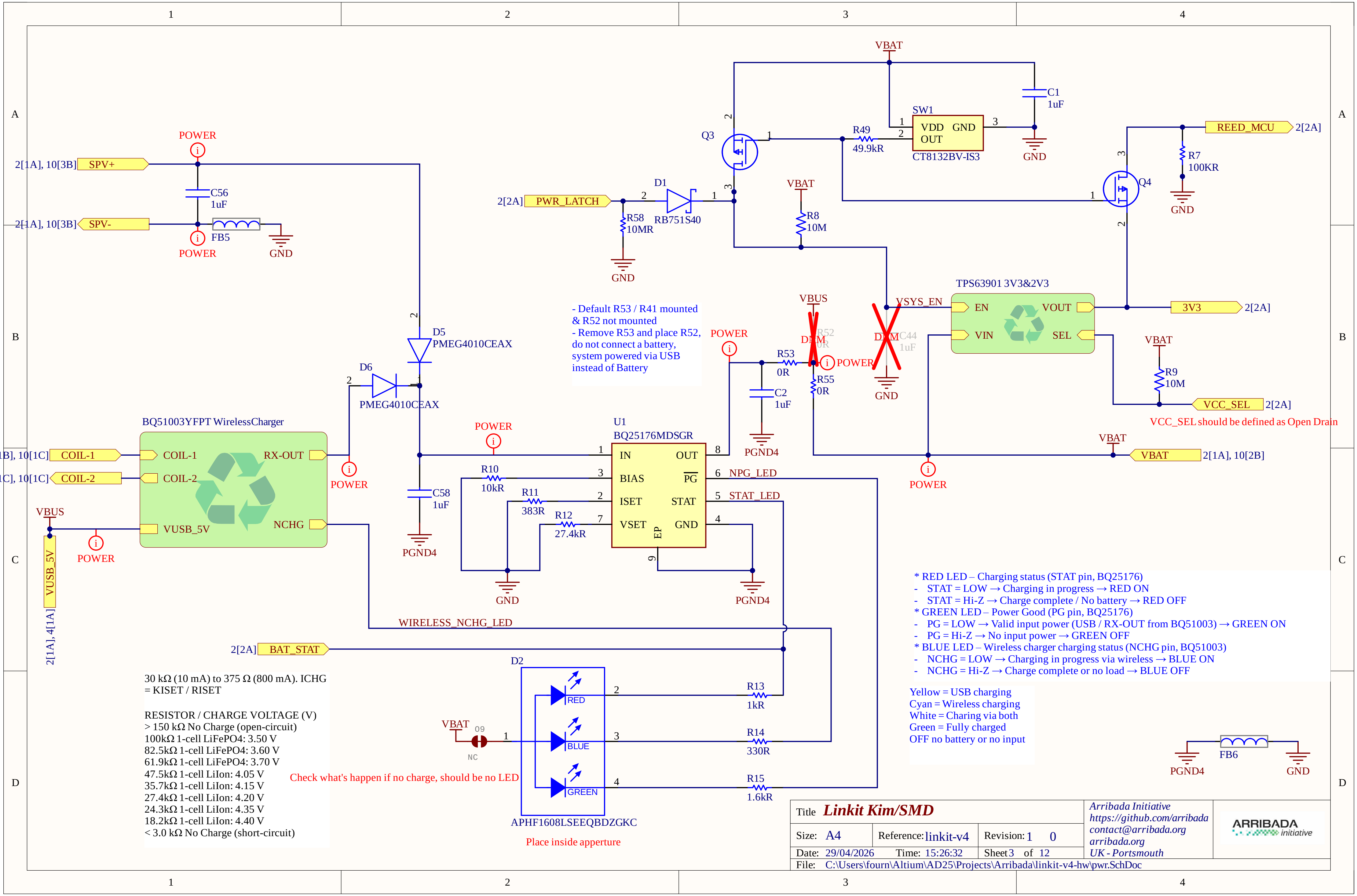
VBAT
C1
1uF
SW1
VDD GND
OUT
CT8132BV-IS3
GND
Q3
R49
49.9kR
REED_MCU
2[2A]
R7
100KR
Q4
GND
POWER
2[1A], 10[3B]
SPV+
C56
1uF
SPV-
FB5
GND
2[2A]
PWR_LATCH
D1
RB751S40
R58
10MR
GND
R8
10M
TPS63901 3V3&2V3
EN
VOUT
VIN
SEL
VSYS_EN
3V3
2[2A]
VBAT
R9
10M
VCC_SEL
2[2A]
VCC_SEL should be defined as Open Drain
- Default R53 / R41 mounted & R52 not mounted
- Remove R53 and place R52, do not connect a battery, system powered via USB instead of Battery
D5
PMEG4010CEAX
D6
PMEG4010CEAX
VBUS
R52
0R
DNM
C44
1uF
R53
0R
C2
1uF
PGND4
R55
0R
BQ51003YFPT WirelessCharger
COIL-1
COIL-2
RX-OUT
VUSB_5V
NCHG
U1
BQ25176MDSGR
IN
OUT
BIAS
PG
ISET
STAT
VSET
GND
EP
NPG_LED
STAT_LED
R10
10kR
R11
383R
R12
27.4kR
C58
1uF
PGND4
GND
PGND4
VBAT
2[1A], 10[2B]
VUSB_5V
2[1A], 4[1A]
WIRELESS_NCHG_LED
2[2A]
BAT_STAT
* RED LED – Charging status (STAT pin, BQ25176)
- STAT = LOW → Charging in progress → RED ON
- STAT = Hi-Z → Charge complete / No battery → RED OFF
* GREEN LED – Power Good (PG pin, BQ25176)
- PG = LOW → Valid input power (USB / RX-OUT from BQ51003) → GREEN ON
- PG = Hi-Z → No input power → GREEN OFF
* BLUE LED – Wireless charger charging status (NCHG pin, BQ51003)
- NCHG = LOW → Charging in progress via wireless → BLUE ON
- NCHG = Hi-Z → Charge complete or no load → BLUE OFF
Yellow = USB charging
Cyan = Wireless charging
White = Charing via both
Green = Fully charged
OFF no battery or no input
30 kΩ (10 mA) to 375 Ω (800 mA). ICHG = KISET / RISET
RESISTOR / CHARGE VOLTAGE (V)
> 150 kΩ No Charge (open-circuit)
100kΩ 1-cell LiFePO4: 3.50 V
82.5kΩ 1-cell LiFePO4: 3.60 V
61.9kΩ 1-cell LiFePO4: 3.70 V
47.5kΩ 1-cell LiIon: 4.05 V
35.7kΩ 1-cell LiIon: 4.15 V
27.4kΩ 1-cell LiIon: 4.20 V
24.3kΩ 1-cell LiIon: 4.35 V
18.2kΩ 1-cell LiIon: 4.40 V
< 3.0 kΩ No Charge (short-circuit)
D2
RED
BLUE
GREEN
VBAT
NC
Check what's happen if no charge, should be no LED
APHF1608LSEEQBDZGKC
Place inside apperture
R13
1kR
R14
330R
R15
1.6kR
FB6
PGND4
GND
Title Linkit Kim/SMD
Size: A4
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Sheet 3 of 12
File: C:\Users\fourn\Altium\AD25\Projects\Arribada\linkit-v4-hw\pwr.SchDoc
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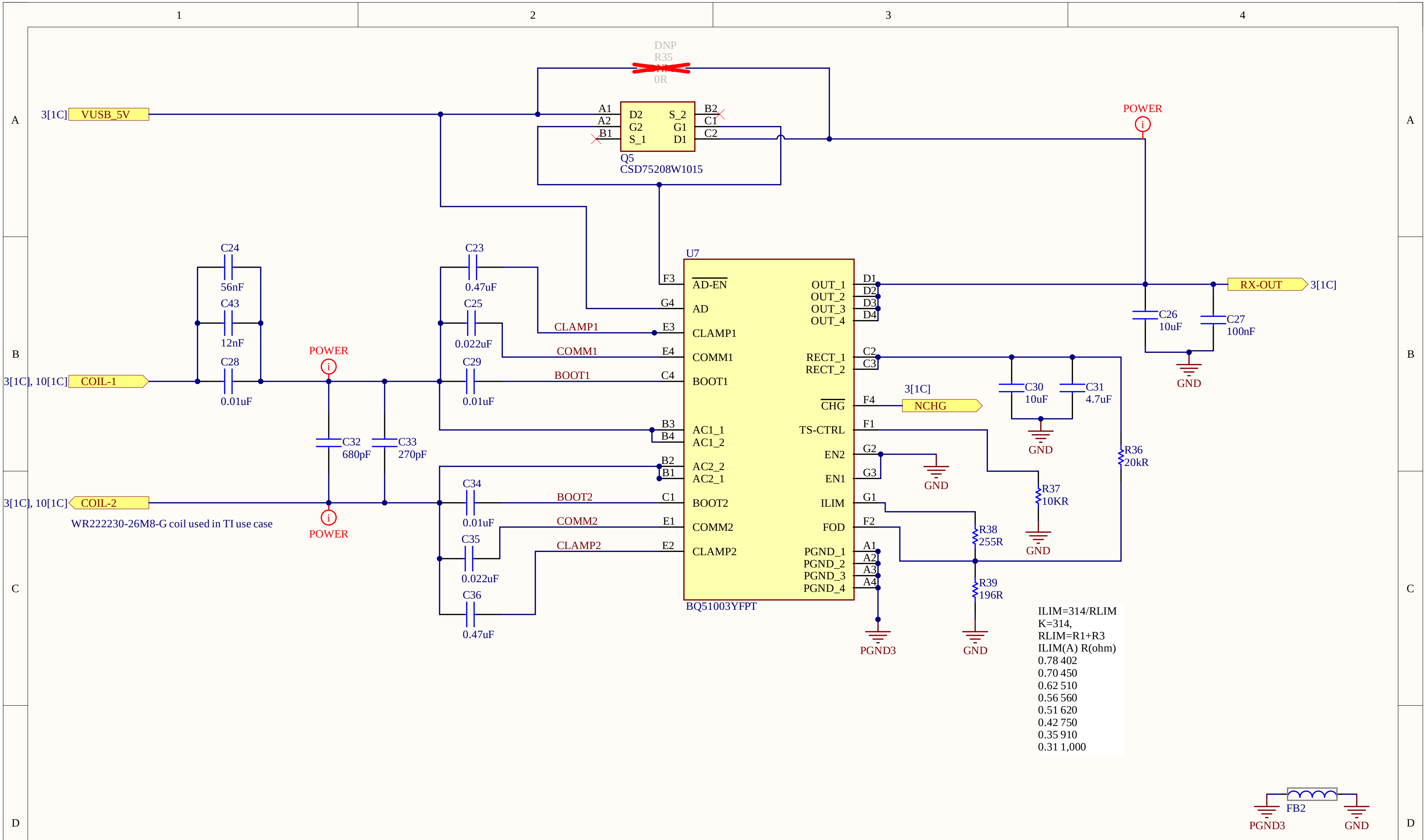


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Date: 29/04/2026 | Time: 15:26:32 | Sheet 4 of 12

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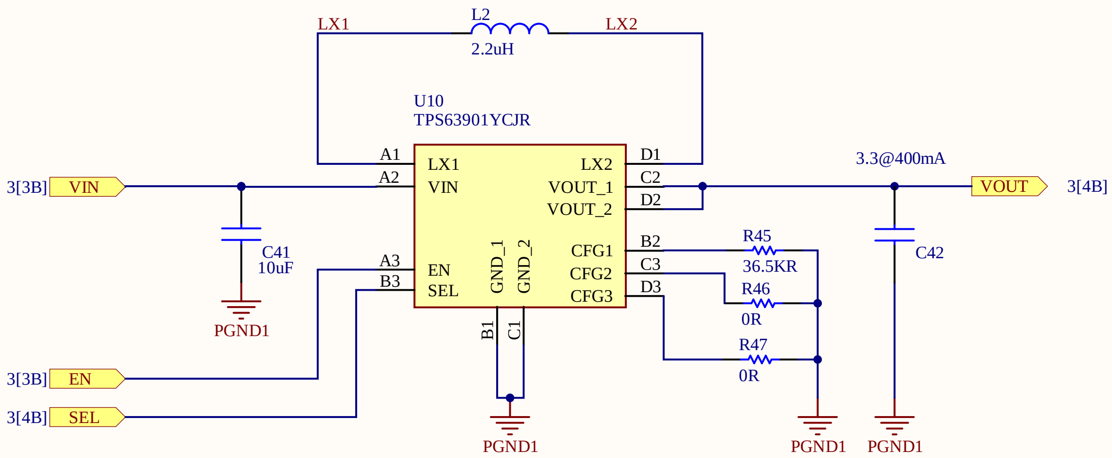
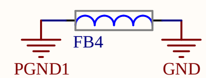



| Output Voltage – $V_{O(2)}$ (SEL = HIGH) | | Input Current Limit | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | UNLIMITED | 100 mA | 50 mA | 25 mA | 10 mA | 5 mA | 2.5 mA | 1 mA |
| 1.8 V | $R_{CFG1}$ | 0 Ω | | | | | | | |
| | $R_{CFG2}$ | 0 Ω | 511 Ω | 1.15 kΩ | 1.87 kΩ | 2.74 kΩ | 3.83 kΩ | 5.11 kΩ | 6.49 kΩ |
| 1.9 V | $R_{CFG1}$ | 511 Ω | | | | | | | |
| | $R_{CFG2}$ | 0 Ω | 511 Ω | 1.15 kΩ | 1.87 kΩ | 2.74 kΩ | 3.83 kΩ | 5.11 kΩ | 6.49 kΩ |
| 2.0 V | $R_{CFG1}$ | 1.15 kΩ | | | | | | | |
| | $R_{CFG2}$ | 0 Ω | 511 Ω | 1.15 kΩ | 1.87 kΩ | 2.74 kΩ | 3.83 kΩ | 5.11 kΩ | 6.49 kΩ |
| 2.1 V | $R_{CFG1}$ | 1.87 kΩ | | | | | | | |
| | $R_{CFG2}$ | 0 Ω | 511 Ω | 1.15 kΩ | 1.87 kΩ | 2.74 kΩ | 3.83 kΩ | 5.11 kΩ | 6.49 kΩ |
| 2.2 V | $R_{CFG1}$ | 2.74 kΩ | | | | | | | |
| | $R_{CFG2}$ | 0 Ω | 511 Ω | 1.15 kΩ | 1.87 kΩ | 2.74 kΩ | 3.83 kΩ | 5.11 kΩ | 6.49 kΩ |
| 2.3 V | $R_{CFG1}$ | 3.83 kΩ | | | | | | | |
| | $R_{CFG2}$ | 0 Ω | 511 Ω | 1.15 kΩ | 1.87 kΩ | 2.74 kΩ | 3.83 kΩ | 5.11 kΩ | 6.49 kΩ |
| 2.4 V | $R_{CFG1}$ | 5.11 kΩ | | | | | | | |
| | $R_{CFG2}$ | 0 Ω | 511 Ω | 1.15 kΩ | 1.87 kΩ | 2.74 kΩ | 3.83 kΩ | 5.11 kΩ | 6.49 kΩ |
| 2.5 V | $R_{CFG1}$ | 6.49 kΩ | | | | | | | |
| | $R_{CFG2}$ | 0 Ω | 511 Ω | 1.15 kΩ | 1.87 kΩ | 2.74 kΩ | 3.83 kΩ | 5.11 kΩ | 6.49 kΩ |
| 2.6 V | $R_{CFG1}$ | 8.25 kΩ | | | | | | | |
| | $R_{CFG2}$ | 0 Ω | 511 Ω | 1.15 kΩ | 1.87 kΩ | 2.74 kΩ | 3.83 kΩ | 5.11 kΩ | 6.49 kΩ |
| 2.7 V | $R_{CFG1}$ | 10.5 kΩ | | | | | | | |
| | $R_{CFG2}$ | 0 Ω | 511 Ω | 1.15 kΩ | 1.87 kΩ | 2.74 kΩ | 3.83 kΩ | 5.11 kΩ | 6.49 kΩ |
| 2.8 V | $R_{CFG1}$ | 13.3 kΩ | | | | | | | |
| | $R_{CFG2}$ | 0 Ω | 511 Ω | 1.15 kΩ | 1.87 kΩ | 2.74 kΩ | 3.83 kΩ | 5.11 kΩ | 6.49 kΩ |
| 2.9 V | $R_{CFG1}$ | 16.2 kΩ | | | | | | | |
| | $R_{CFG2}$ | 0 Ω | 511 Ω | 1.15 kΩ | 1.87 kΩ | 2.74 kΩ | 3.83 kΩ | 5.11 kΩ | 6.49 kΩ |
| 3.0 V | $R_{CFG1}$ | 20.5 kΩ | | | | | | | |
| | $R_{CFG2}$ | 0 Ω | 511 Ω | 1.15 kΩ | 1.87 kΩ | 2.74 kΩ | 3.83 kΩ | 5.11 kΩ | 6.49 kΩ |
| 3.1 V | $R_{CFG1}$ | 24.9 kΩ | | | | | | | |
| | $R_{CFG2}$ | 0 Ω | 511 Ω | 1.15 kΩ | 1.87 kΩ | 2.74 kΩ | 3.83 kΩ | 5.11 kΩ | 6.49 kΩ |
| 3.2 V | $R_{CFG1}$ | 30.1 kΩ | | | | | | | |
| | $R_{CFG2}$ | 0 Ω | 511 Ω | 1.15 kΩ | 1.87 kΩ | 2.74 kΩ | 3.83 kΩ | 5.11 kΩ | 6.49 kΩ |
| 3.3 V | $R_{CFG1}$ | 36.5 kΩ | | | | | | | |
| | $R_{CFG2}$ | 0 Ω | 511 Ω | 1.15 kΩ | 1.87 kΩ | 2.74 kΩ | 3.83 kΩ | 5.11 kΩ | 6.49 kΩ |
| 3.4 V | $R_{CFG1}$ | 0 Ω | | | | | | | |
| | $R_{CFG2}$ | 8.25 kΩ | 10.5 kΩ | 13.3 kΩ | 16.2 kΩ | 20.5 kΩ | 24.9 kΩ | 30.1 kΩ | 36.5 kΩ |
| 3.5 V | $R_{CFG1}$ | 511 Ω | | | | | | | |
| | $R_{CFG2}$ | 8.25 kΩ | 10.5 kΩ | 13.3 kΩ | 16.2 kΩ | 20.5 kΩ | 24.9 kΩ | 30.1 kΩ | 36.5 kΩ |
| 3.6 V | $R_{CFG1}$ | 1.15 kΩ | | | | | | | |
| | $R_{CFG2}$ | 8.25 kΩ | 10.5 kΩ | 13.3 kΩ | 16.2 kΩ | 20.5 kΩ | 24.9 kΩ | 30.1 kΩ | 36.5 kΩ |
| 3.7 V | $R_{CFG1}$ | 1.87 kΩ | | | | | | | |
| | $R_{CFG2}$ | 8.25 kΩ | 10.5 kΩ | 13.3 kΩ | 16.2 kΩ | 20.5 kΩ | 24.9 kΩ | 30.1 kΩ | 36.5 kΩ |
| 3.8 V | $R_{CFG1}$ | 2.74 kΩ | | | | | | | |
| | $R_{CFG2}$ | 8.25 kΩ | 10.5 kΩ | 13.3 kΩ | 16.2 kΩ | 20.5 kΩ | 24.9 kΩ | 30.1 kΩ | 36.5 kΩ |
| 3.9 V | $R_{CFG1}$ | 3.83 kΩ | | | | | | | |
| | $R_{CFG2}$ | 8.25 kΩ | 10.5 kΩ | 13.3 kΩ | 16.2 kΩ | 20.5 kΩ | 24.9 kΩ | 30.1 kΩ | 36.5 kΩ |
| 4.0 V | $R_{CFG1}$ | 5.11 kΩ | | | | | | | |
| | $R_{CFG2}$ | 8.25 kΩ | 10.5 kΩ | 13.3 kΩ | 16.2 kΩ | 20.5 kΩ | 24.9 kΩ | 30.1 kΩ | 36.5 kΩ |
| 4.1 V | $R_{CFG1}$ | 6.49 kΩ | | | | | | | |
| | $R_{CFG2}$ | 8.25 kΩ | 10.5 kΩ | 13.3 kΩ | 16.2 kΩ | 20.5 kΩ | 24.9 kΩ | 30.1 kΩ | 36.5 kΩ |
| 4.2 V | $R_{CFG1}$ | 8.25 kΩ | | | | | | | |
| | $R_{CFG2}$ | 8.25 kΩ | 10.5 kΩ | 13.3 kΩ | 16.2 kΩ | 20.5 kΩ | 24.9 kΩ | 30.1 kΩ | 36.5 kΩ |
| 4.3 V | $R_{CFG1}$ | 10.5 kΩ | | | | | | | |
| | $R_{CFG2}$ | 8.25 kΩ | 10.5 kΩ | 13.3 kΩ | 16.2 kΩ | 20.5 kΩ | 24.9 kΩ | 30.1 kΩ | 36.5 kΩ |
| 4.4 V | $R_{CFG1}$ | 13.3 kΩ | | | | | | | |
| | $R_{CFG2}$ | 8.25 kΩ | 10.5 kΩ | 13.3 kΩ | 16.2 kΩ | 20.5 kΩ | 24.9 kΩ | 30.1 kΩ | 36.5 kΩ |
| 4.5 V | $R_{CFG1}$ | 16.2 kΩ | | | | | | | |
| | $R_{CFG2}$ | 8.25 kΩ | 10.5 kΩ | 13.3 kΩ | 16.2 kΩ | 20.5 kΩ | 24.9 kΩ | 30.1 kΩ | 36.5 kΩ |
| 4.6 V | $R_{CFG1}$ | 20.5 kΩ | | | | | | | |
| | $R_{CFG2}$ | 8.25 kΩ | 10.5 kΩ | 13.3 kΩ | 16.2 kΩ | 20.5 kΩ | 24.9 kΩ | 30.1 kΩ | 36.5 kΩ |
| 4.7 V | $R_{CFG1}$ | 24.9 kΩ | | | | | | | |
| | $R_{CFG2}$ | 8.25 kΩ | 10.5 kΩ | 13.3 kΩ | 16.2 kΩ | 20.5 kΩ | 24.9 kΩ | 30.1 kΩ | 36.5 kΩ |
| 4.8 V | $R_{CFG1}$ | 30.1 kΩ | | | | | | | |
| | $R_{CFG2}$ | 8.25 kΩ | 10.5 kΩ | 13.3 kΩ | 16.2 kΩ | 20.5 kΩ | 24.9 kΩ | 30.1 kΩ | 36.5 kΩ |
| 5.0 V | $R_{CFG1}$ | 36.5 kΩ | | | | | | | |
| | $R_{CFG2}$ | 8.25 kΩ | 10.5 kΩ | 13.3 kΩ | 16.2 kΩ | 20.5 kΩ | 24.9 kΩ | 30.1 kΩ | 36.5 kΩ |

**Table 6-3. Output Voltage (SEL Pin = Low) Settings**

| Output Voltage – $V_{O(1)}$ (SEL = LOW) | $R_{CFG3}$ |
|---|---|
| 1.8 V | 0 Ω |
| 2.0 V | 511 Ω |
| 2.1 V | 1.15 kΩ |
| 2.2 V | 1.87 kΩ |
| 2.3 V | 2.74 kΩ |
| 2.4 V | 3.83 kΩ |
| 2.5 V | 5.11 kΩ |
| 2.6 V | 6.49 kΩ |
| 2.7 V | 8.25 kΩ |
| 2.8 V | 10.5 kΩ |
| 3.0 V | 13.3 kΩ |
| 3.3 V | 16.2 kΩ |
| 3.6 V | 20.5 kΩ |
| 4.0 V | 24.9 kΩ |
| 4.5 V | 30.1 kΩ |
| 5.0 V | 36.5 kΩ |




Title *Linkit Kim/SMD*

Size: A4 Reference: linkit-v4 Revision: 1 0

Date: 29/04/2026 Time: 15:26:32 Sheet 5 of 12

File: C:\Users\fourn\Altium\AD25\Projects\Arribada\hardware-lib\DeviceSheets\Power\TPS63901_BuckBoost_lowIq_3V3&2V3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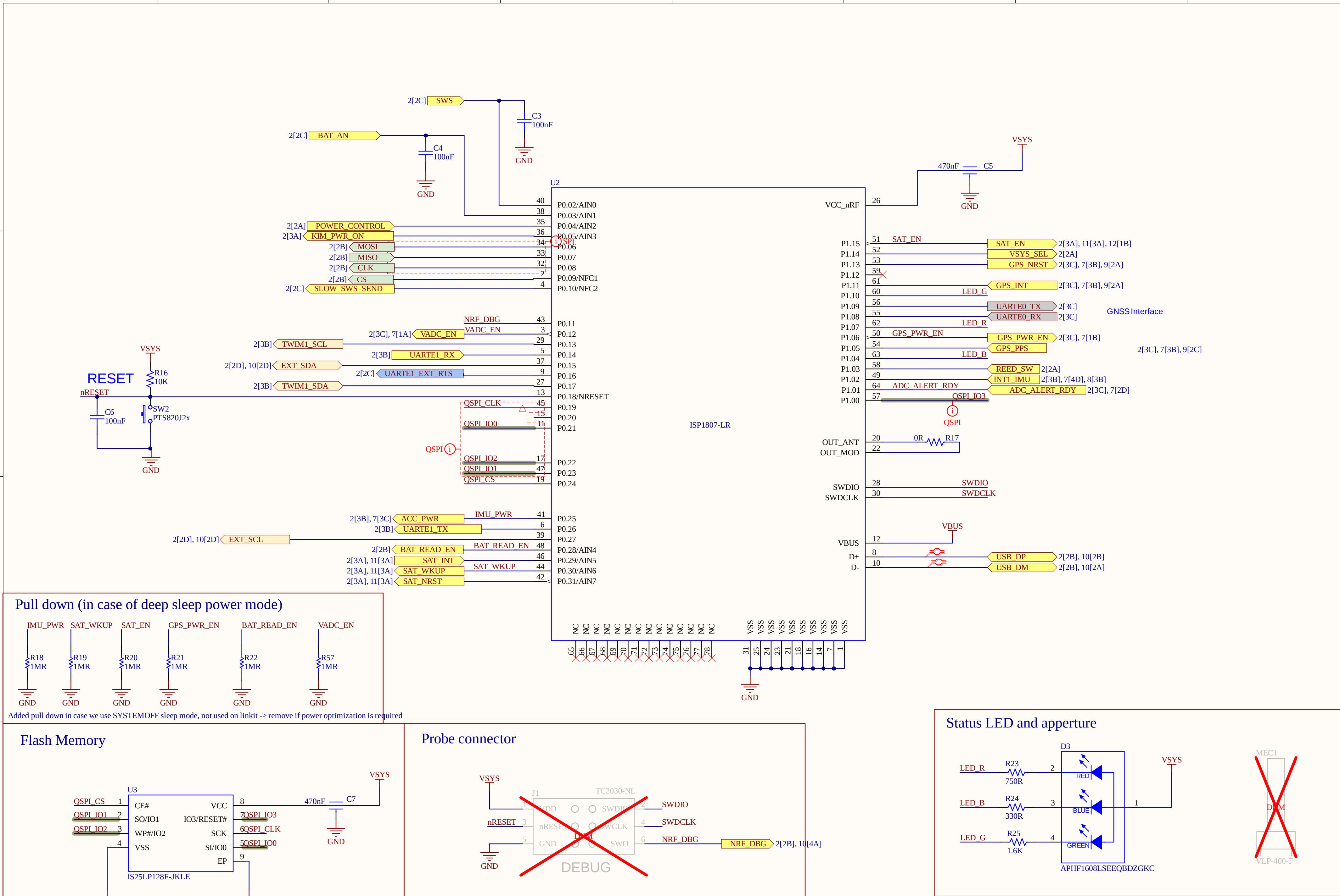


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Date: 29/04/2026 | Time: 15:26:33 | Sheet 6 of 12

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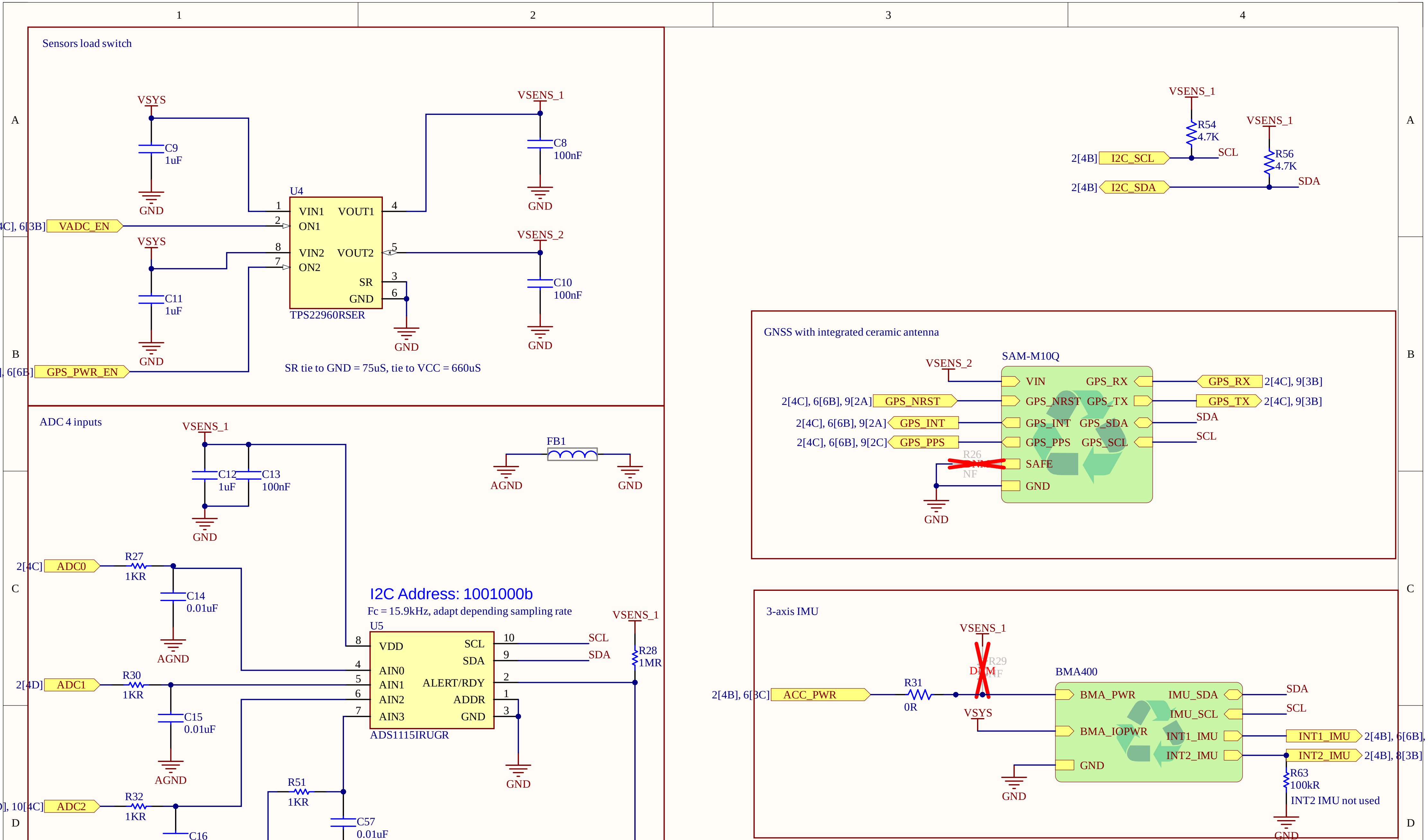


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| Title | *Linkit Kim/SMD* | | |
| Size: A4 | Reference: linkit-v4 | Revision: 1 0 | |
| Date: 29/04/2026 | Time: 15:26:34 | Sheet 7 of 12 | |
| File: C:\Users\fourn\Altium\AD25\Projects\Arribada\linkit-v4-hw\sensors.SchDoc | | | |

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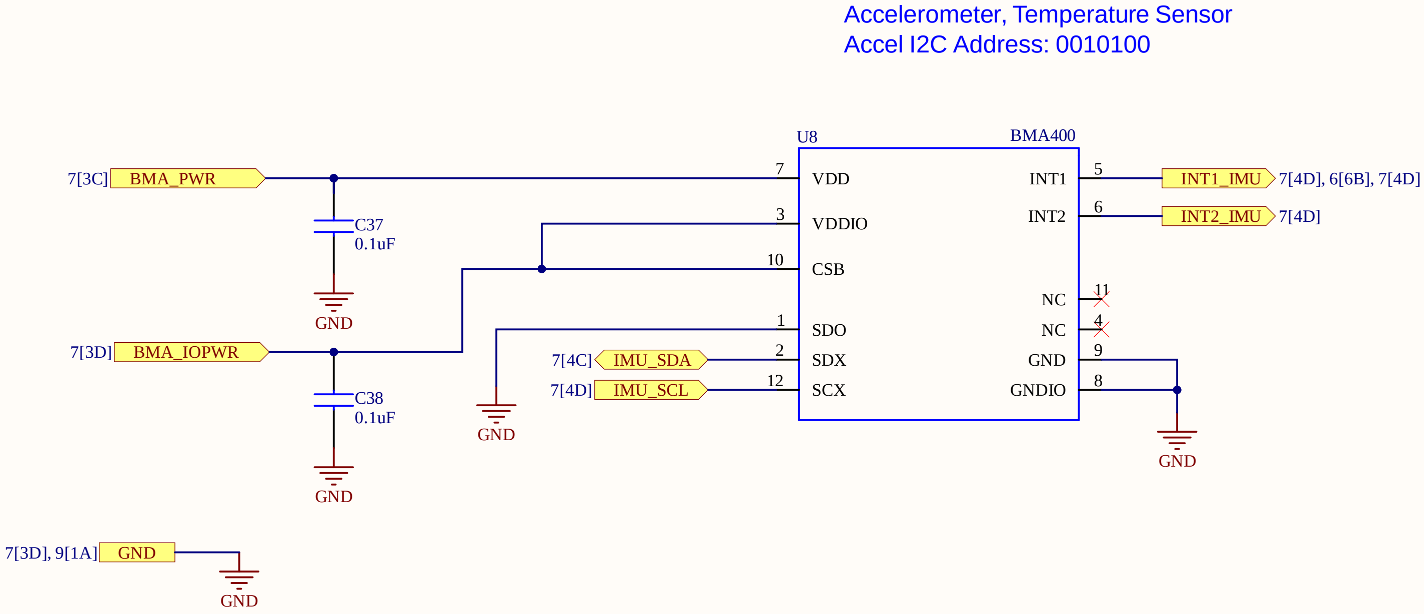


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Date: 29/04/2026 Time: 15:26:34 Sheet of

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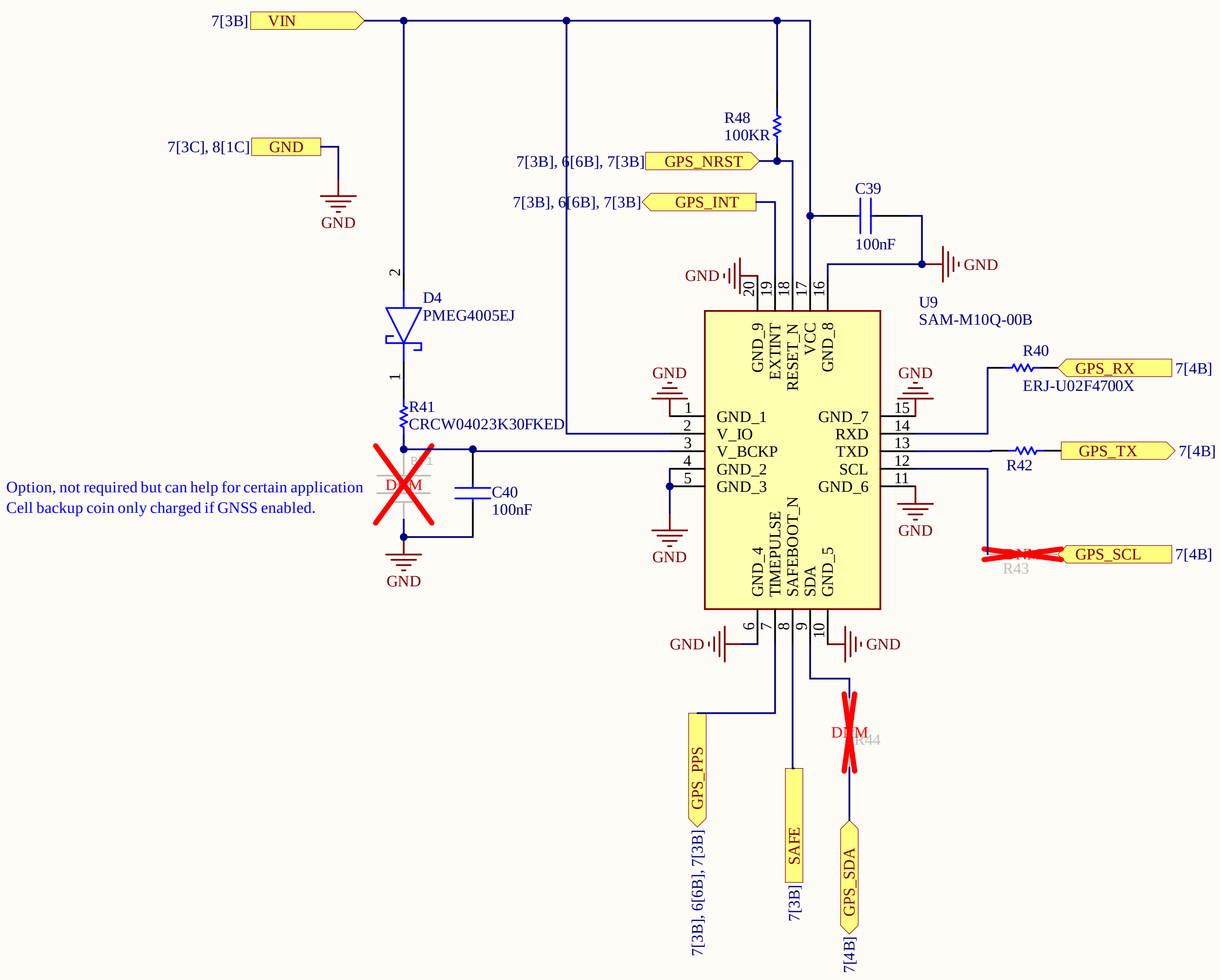


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|---|---|---|---|
| Title **Linkit Kim/SMD** | | | *Arribada Initiative* |
| Size: A4 | Reference: linkit-v4 | Revision: 1 0 | *https://github.com/arribada* |
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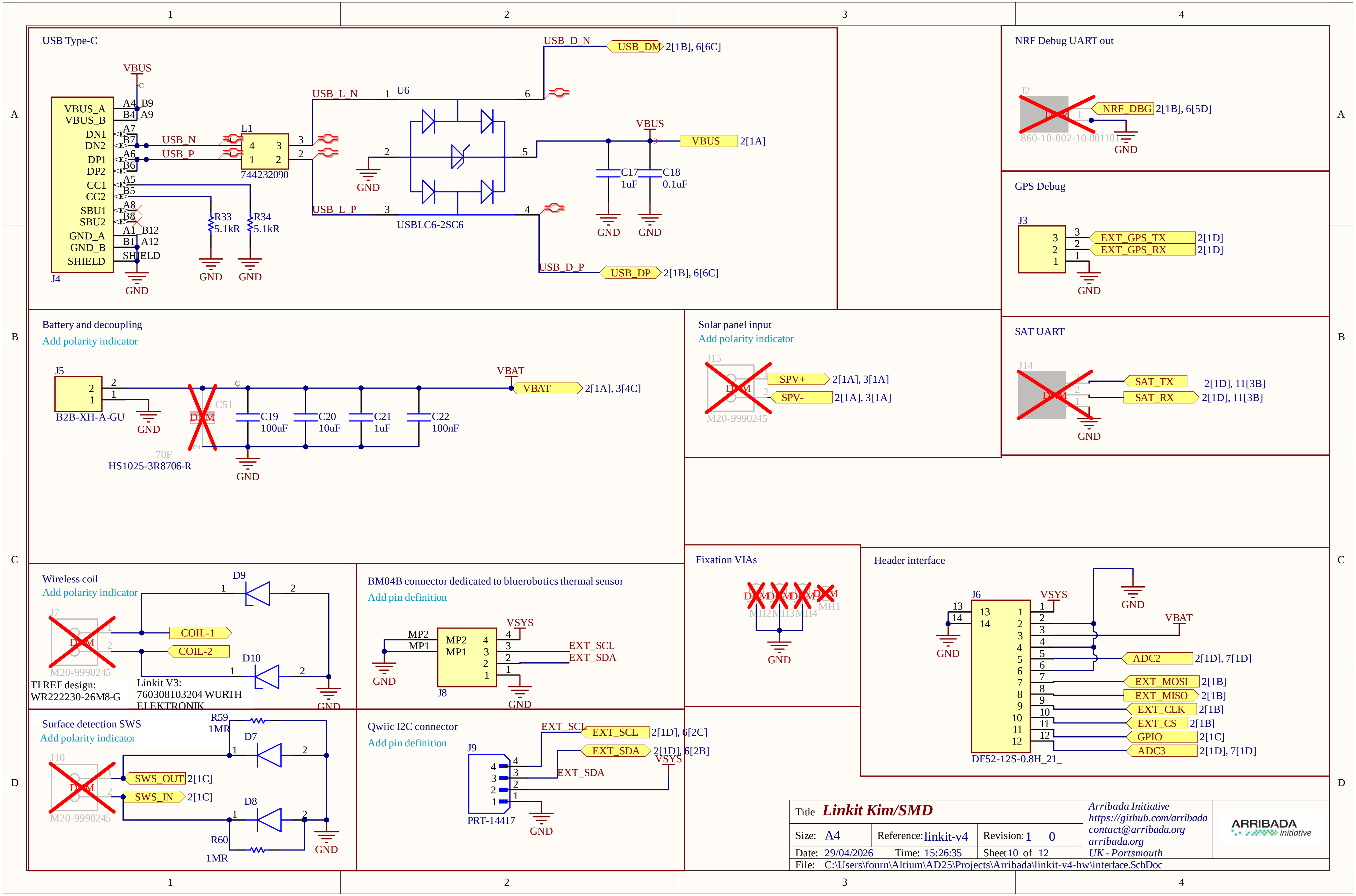
File: C:\Users\fourn\Altium\AD25\Projects\Arribada\hardware-lib\DeviceSheets\GPS\SAM-M10Q.SchDoc

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Size: A4 | Reference: linkit-v4 | Revision: 1 0

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File: C:\Users\fourn\Altium\AD25\Projects\Arribada\linkit-v4-hw\interface.SchDoc

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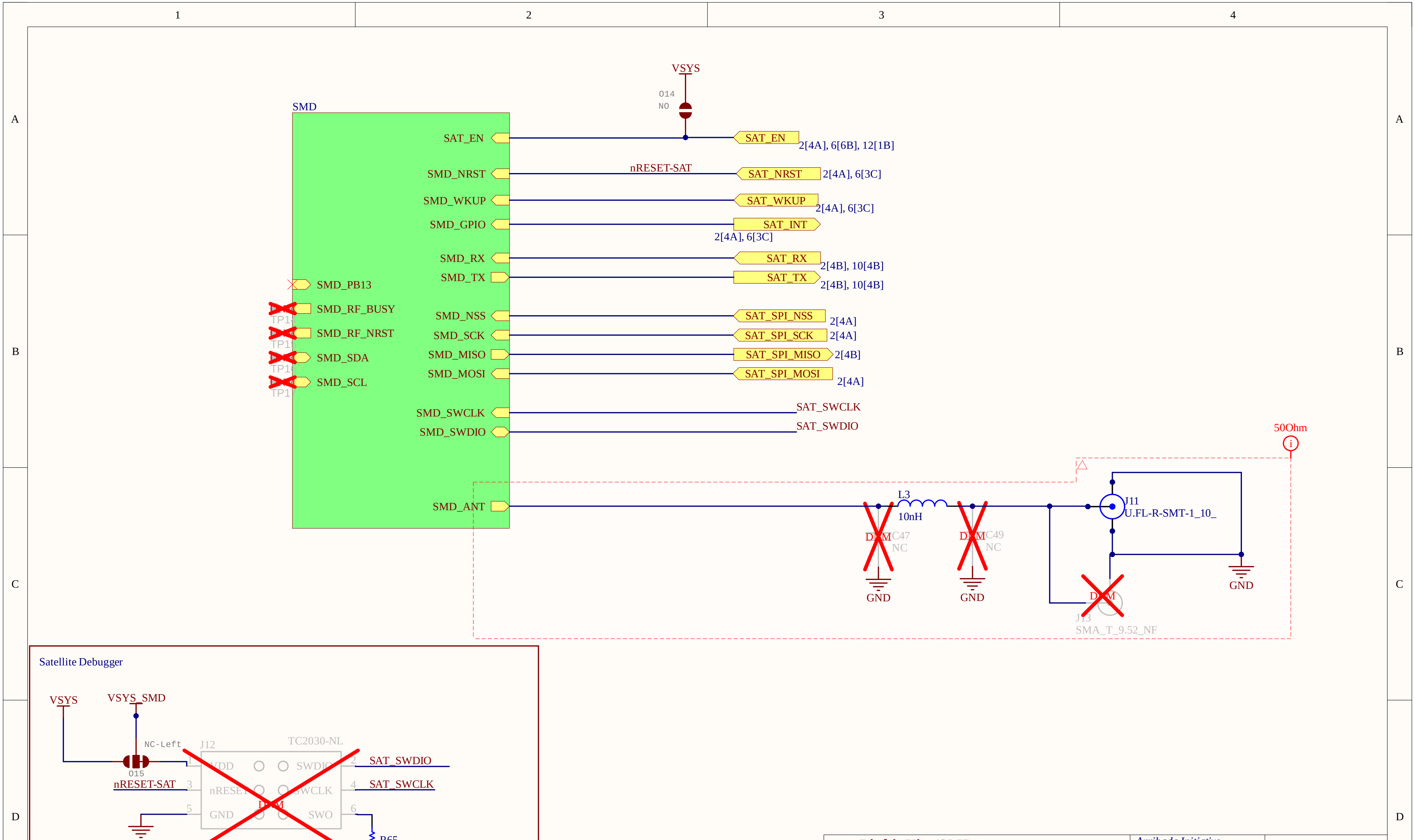


| | |
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| Title | *Linkit Kim/SMD* |
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| Revision: 1 0 | |
| Date: 29/04/2026 | Time: 15:26:36 |
| Sheet 11 of 12 | |
| File: | C:\Users\fourn\Altium\AD25\Projects\Arribada\linkit-v4-hw\satellite.SchDoc |

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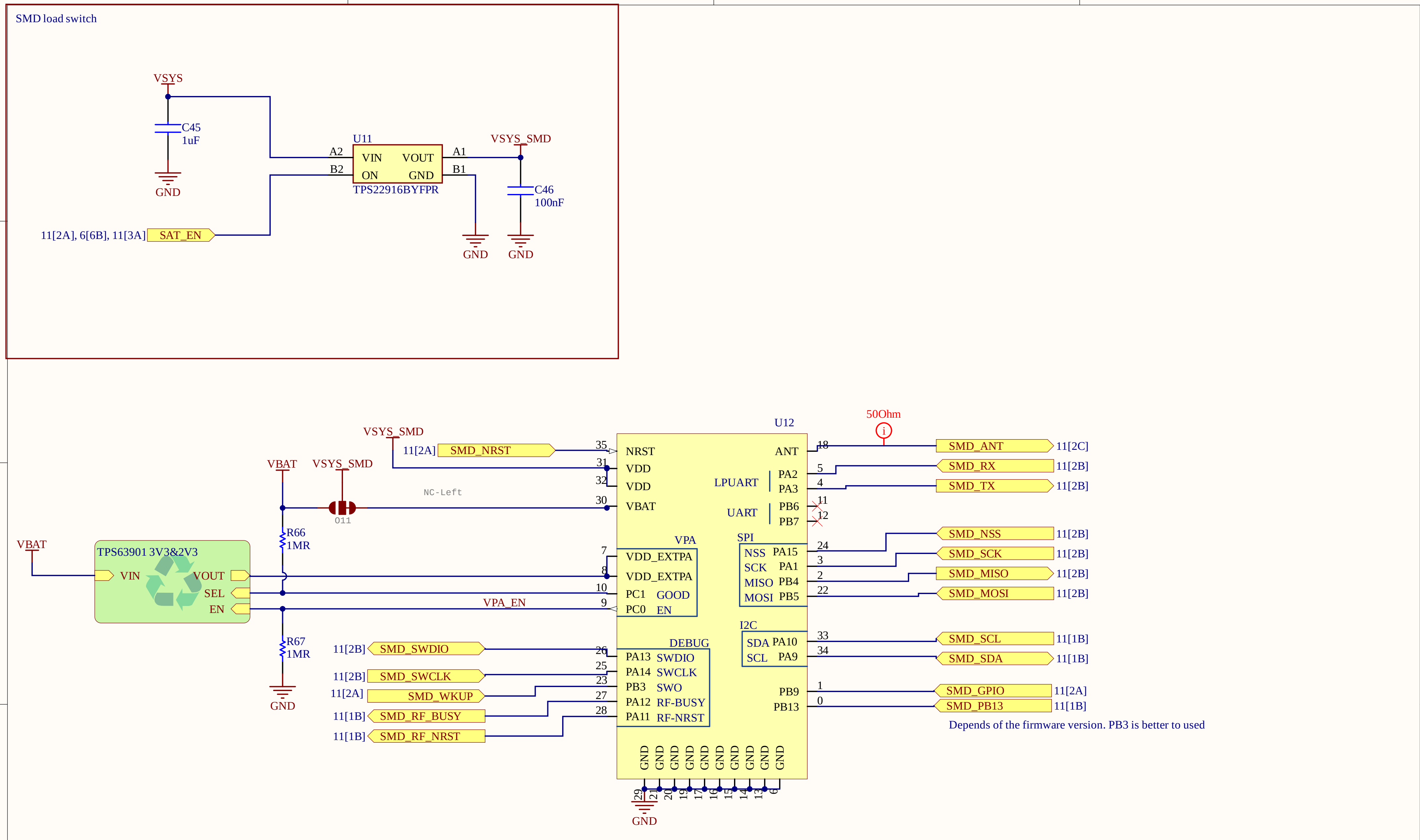


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Date: 29/04/2026 | Time: 15:26:36 | Sheet 12 of 12

File: C:\Users\fourn\Altium\AD25\Projects\Arribada\linkit-v4-hw\sat-smd.SchDoc

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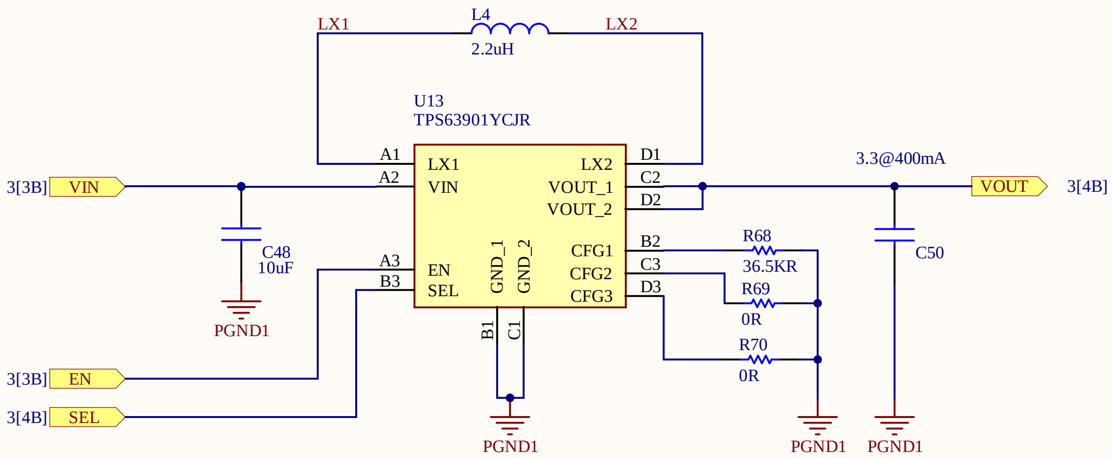
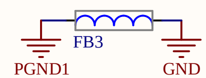



| Output Voltage – $V_{O(2)}$ (SEL = HIGH) | | Input Current Limit | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | UNLIMITED | 100 mA | 50 mA | 25 mA | 10 mA | 5 mA | 2.5 mA | 1 mA |
| 1.8 V | $R_{CFG1}$ | 0 Ω | | | | | | | |
| | $R_{CFG2}$ | 0 Ω | 511 Ω | 1.15 kΩ | 1.87 kΩ | 2.74 kΩ | 3.83 kΩ | 5.11 kΩ | 6.49 kΩ |
| 1.9 V | $R_{CFG1}$ | 511 Ω | | | | | | | |
| | $R_{CFG2}$ | 0 Ω | 511 Ω | 1.15 kΩ | 1.87 kΩ | 2.74 kΩ | 3.83 kΩ | 5.11 kΩ | 6.49 kΩ |
| 2.0 V | $R_{CFG1}$ | 1.15 kΩ | | | | | | | |
| | $R_{CFG2}$ | 0 Ω | 511 Ω | 1.15 kΩ | 1.87 kΩ | 2.74 kΩ | 3.83 kΩ | 5.11 kΩ | 6.49 kΩ |
| 2.1 V | $R_{CFG1}$ | 1.87 kΩ | | | | | | | |
| | $R_{CFG2}$ | 0 Ω | 511 Ω | 1.15 kΩ | 1.87 kΩ | 2.74 kΩ | 3.83 kΩ | 5.11 kΩ | 6.49 kΩ |
| 2.2 V | $R_{CFG1}$ | 2.74 kΩ | | | | | | | |
| | $R_{CFG2}$ | 0 Ω | 511 Ω | 1.15 kΩ | 1.87 kΩ | 2.74 kΩ | 3.83 kΩ | 5.11 kΩ | 6.49 kΩ |
| 2.3 V | $R_{CFG1}$ | 3.83 kΩ | | | | | | | |
| | $R_{CFG2}$ | 0 Ω | 511 Ω | 1.15 kΩ | 1.87 kΩ | 2.74 kΩ | 3.83 kΩ | 5.11 kΩ | 6.49 kΩ |
| 2.4 V | $R_{CFG1}$ | 5.11 kΩ | | | | | | | |
| | $R_{CFG2}$ | 0 Ω | 511 Ω | 1.15 kΩ | 1.87 kΩ | 2.74 kΩ | 3.83 kΩ | 5.11 kΩ | 6.49 kΩ |
| 2.5 V | $R_{CFG1}$ | 6.49 kΩ | | | | | | | |
| | $R_{CFG2}$ | 0 Ω | 511 Ω | 1.15 kΩ | 1.87 kΩ | 2.74 kΩ | 3.83 kΩ | 5.11 kΩ | 6.49 kΩ |
| 2.6 V | $R_{CFG1}$ | 8.25 kΩ | | | | | | | |
| | $R_{CFG2}$ | 0 Ω | 511 Ω | 1.15 kΩ | 1.87 kΩ | 2.74 kΩ | 3.83 kΩ | 5.11 kΩ | 6.49 kΩ |
| 2.7 V | $R_{CFG1}$ | 10.5 kΩ | | | | | | | |
| | $R_{CFG2}$ | 0 Ω | 511 Ω | 1.15 kΩ | 1.87 kΩ | 2.74 kΩ | 3.83 kΩ | 5.11 kΩ | 6.49 kΩ |
| 2.8 V | $R_{CFG1}$ | 13.3 kΩ | | | | | | | |
| | $R_{CFG2}$ | 0 Ω | 511 Ω | 1.15 kΩ | 1.87 kΩ | 2.74 kΩ | 3.83 kΩ | 5.11 kΩ | 6.49 kΩ |
| 2.9 V | $R_{CFG1}$ | 16.2 kΩ | | | | | | | |
| | $R_{CFG2}$ | 0 Ω | 511 Ω | 1.15 kΩ | 1.87 kΩ | 2.74 kΩ | 3.83 kΩ | 5.11 kΩ | 6.49 kΩ |
| 3.0 V | $R_{CFG1}$ | 20.5 kΩ | | | | | | | |
| | $R_{CFG2}$ | 0 Ω | 511 Ω | 1.15 kΩ | 1.87 kΩ | 2.74 kΩ | 3.83 kΩ | 5.11 kΩ | 6.49 kΩ |
| 3.1 V | $R_{CFG1}$ | 24.9 kΩ | | | | | | | |
| | $R_{CFG2}$ | 0 Ω | 511 Ω | 1.15 kΩ | 1.87 kΩ | 2.74 kΩ | 3.83 kΩ | 5.11 kΩ | 6.49 kΩ |
| 3.2 V | $R_{CFG1}$ | 30.1 kΩ | | | | | | | |
| | $R_{CFG2}$ | 0 Ω | 511 Ω | 1.15 kΩ | 1.87 kΩ | 2.74 kΩ | 3.83 kΩ | 5.11 kΩ | 6.49 kΩ |
| 3.3 V | $R_{CFG1}$ | 36.5 kΩ | | | | | | | |
| | $R_{CFG2}$ | 0 Ω | 511 Ω | 1.15 kΩ | 1.87 kΩ | 2.74 kΩ | 3.83 kΩ | 5.11 kΩ | 6.49 kΩ |
| 3.4 V | $R_{CFG1}$ | 0 Ω | | | | | | | |
| | $R_{CFG2}$ | 8.25 kΩ | 10.5 kΩ | 13.3 kΩ | 16.2 kΩ | 20.5 kΩ | 24.9 kΩ | 30.1 kΩ | 36.5 kΩ |
| 3.5 V | $R_{CFG1}$ | 511 Ω | | | | | | | |
| | $R_{CFG2}$ | 8.25 kΩ | 10.5 kΩ | 13.3 kΩ | 16.2 kΩ | 20.5 kΩ | 24.9 kΩ | 30.1 kΩ | 36.5 kΩ |
| 3.6 V | $R_{CFG1}$ | 1.15 kΩ | | | | | | | |
| | $R_{CFG2}$ | 8.25 kΩ | 10.5 kΩ | 13.3 kΩ | 16.2 kΩ | 20.5 kΩ | 24.9 kΩ | 30.1 kΩ | 36.5 kΩ |
| 3.7 V | $R_{CFG1}$ | 1.87 kΩ | | | | | | | |
| | $R_{CFG2}$ | 8.25 kΩ | 10.5 kΩ | 13.3 kΩ | 16.2 kΩ | 20.5 kΩ | 24.9 kΩ | 30.1 kΩ | 36.5 kΩ |
| 3.8 V | $R_{CFG1}$ | 2.74 kΩ | | | | | | | |
| | $R_{CFG2}$ | 8.25 kΩ | 10.5 kΩ | 13.3 kΩ | 16.2 kΩ | 20.5 kΩ | 24.9 kΩ | 30.1 kΩ | 36.5 kΩ |
| 3.9 V | $R_{CFG1}$ | 3.83 kΩ | | | | | | | |
| | $R_{CFG2}$ | 8.25 kΩ | 10.5 kΩ | 13.3 kΩ | 16.2 kΩ | 20.5 kΩ | 24.9 kΩ | 30.1 kΩ | 36.5 kΩ |
| 4.0 V | $R_{CFG1}$ | 5.11 kΩ | | | | | | | |
| | $R_{CFG2}$ | 8.25 kΩ | 10.5 kΩ | 13.3 kΩ | 16.2 kΩ | 20.5 kΩ | 24.9 kΩ | 30.1 kΩ | 36.5 kΩ |
| 4.1 V | $R_{CFG1}$ | 6.49 kΩ | | | | | | | |
| | $R_{CFG2}$ | 8.25 kΩ | 10.5 kΩ | 13.3 kΩ | 16.2 kΩ | 20.5 kΩ | 24.9 kΩ | 30.1 kΩ | 36.5 kΩ |
| 4.2 V | $R_{CFG1}$ | 8.25 kΩ | | | | | | | |
| | $R_{CFG2}$ | 8.25 kΩ | 10.5 kΩ | 13.3 kΩ | 16.2 kΩ | 20.5 kΩ | 24.9 kΩ | 30.1 kΩ | 36.5 kΩ |
| 4.3 V | $R_{CFG1}$ | 10.5 kΩ | | | | | | | |
| | $R_{CFG2}$ | 8.25 kΩ | 10.5 kΩ | 13.3 kΩ | 16.2 kΩ | 20.5 kΩ | 24.9 kΩ | 30.1 kΩ | 36.5 kΩ |
| 4.4 V | $R_{CFG1}$ | 13.3 kΩ | | | | | | | |
| | $R_{CFG2}$ | 8.25 kΩ | 10.5 kΩ | 13.3 kΩ | 16.2 kΩ | 20.5 kΩ | 24.9 kΩ | 30.1 kΩ | 36.5 kΩ |
| 4.5 V | $R_{CFG1}$ | 16.2 kΩ | | | | | | | |
| | $R_{CFG2}$ | 8.25 kΩ | 10.5 kΩ | 13.3 kΩ | 16.2 kΩ | 20.5 kΩ | 24.9 kΩ | 30.1 kΩ | 36.5 kΩ |
| 4.6 V | $R_{CFG1}$ | 20.5 kΩ | | | | | | | |
| | $R_{CFG2}$ | 8.25 kΩ | 10.5 kΩ | 13.3 kΩ | 16.2 kΩ | 20.5 kΩ | 24.9 kΩ | 30.1 kΩ | 36.5 kΩ |
| 4.7 V | $R_{CFG1}$ | 24.9 kΩ | | | | | | | |
| | $R_{CFG2}$ | 8.25 kΩ | 10.5 kΩ | 13.3 kΩ | 16.2 kΩ | 20.5 kΩ | 24.9 kΩ | 30.1 kΩ | 36.5 kΩ |
| 4.8 V | $R_{CFG1}$ | 30.1 kΩ | | | | | | | |
| | $R_{CFG2}$ | 8.25 kΩ | 10.5 kΩ | 13.3 kΩ | 16.2 kΩ | 20.5 kΩ | 24.9 kΩ | 30.1 kΩ | 36.5 kΩ |
| 5.0 V | $R_{CFG1}$ | 36.5 kΩ | | | | | | | |
| | $R_{CFG2}$ | 8.25 kΩ | 10.5 kΩ | 13.3 kΩ | 16.2 kΩ | 20.5 kΩ | 24.9 kΩ | 30.1 kΩ | 36.5 kΩ |

**Table 6-3. Output Voltage (SEL Pin = Low) Settings**

| Output Voltage – $V_{O(1)}$ (SEL = LOW) | $R_{CFG3}$ |
|---|---|
| 1.8 V | 0 Ω |
| 2.0 V | 511 Ω |
| 2.1 V | 1.15 kΩ |
| 2.2 V | 1.87 kΩ |
| 2.3 V | 2.74 kΩ |
| 2.4 V | 3.83 kΩ |
| 2.5 V | 5.11 kΩ |
| 2.6 V | 6.49 kΩ |
| 2.7 V | 8.25 kΩ |
| 2.8 V | 10.5 kΩ |
| 3.0 V | 13.3 kΩ |
| 3.3 V | 16.2 kΩ |
| 3.6 V | 20.5 kΩ |
| 4.0 V | 24.9 kΩ |
| 4.5 V | 30.1 kΩ |
| 5.0 V | 36.5 kΩ |

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|---|---|---|---|
| Title *Linkit Kim/SMD* | | | *Arribada Initiative* *https://github.com/arribada* *contact@arribada.org* *arribada.org* *UK - Portsmouth* |
| Size: A4 | Reference: linkit-v4 | Revision: 1 0 | |
| Date: 29/04/2026 | Time: 15:26:36 | Sheet 5 of 12 | |
| File: C:\Users\fourn\Altium\AD25\Projects\Arribada\hardware-lib\DeviceSheets\Power\TPS63901_BuckBoost_lowIq_3V3&2V3 | | | |